

COMP 3400 Mini Project Report

Predicting Precipitation from London Weather Data

Baseline Model

I loaded the london_weather.csv dataset containing 15,341 rows and 10 columns. For data cleaning, I filled snow_depth NaN values with 0, justified by the fact that NaN values were concentrated in warmer months and 99% of non-null values were already 0. The date column was dropped and remaining NaN rows were removed using dropna(). The data was split using train_test_split with test_size=0.2 and random_state=42. The baseline Linear Regression model achieved $R^2 = 0.1515$.

Phase 2 — Transformation

Attempt 1 — Min-Max Scaling: I scaled all features to [0,1] using training min and max values. $R^2 = 0.1515$. No change because min-max scaling is a linear transformation and does not affect the predictions of Linear Regression.

Attempt 2 — Standardization (Z-score): I standardized all features to mean=0 and std=1 using training statistics only. $R^2 = 0.1515$. Same result for the same reason — standardization is also a linear transformation.

Attempt 3 — Decimal Scaling: I divided each feature by 10^d where d is the smallest integer making the maximum absolute value fall below 1, computed from training data only. $R^2 = 0.1515$. Again no change as decimal scaling is also a linear transformation. I carried forward the standardized dataset (X_train_z, X_test_z) for subsequent phases.

Phase 3 — Feature Selection

I computed the Pearson correlation matrix on the training data and visualized it using a heatmap. I identified highly correlated feature pairs: mean_temp/min_temp (0.96), max_temp/mean_temp (0.91), max_temp/min_temp (0.81), and sunshine/global_radiation (0.85).

Attempt 1 — Drop min_temp, max_temp, sunshine (threshold > 0.8): $R^2 = 0.1480$. Performance decreased, likely because sunshine has a meaningful correlation with precipitation (-0.24) and removing it lost useful signal.

Attempt 2 — Drop min_temp, max_temp (threshold > 0.9): $R^2 = 0.1489$. Slight improvement over Attempt 1 but still below baseline. Keeping sunshine helped but the temperature redundancy removal was not sufficient to improve performance.

Attempt 3 — Drop snow_depth (lowest correlation with precipitation = 0.00): $R^2 = 0.1515$. No change. snow_depth had near-zero correlation with precipitation so its removal had no effect. No feature selection attempt improved upon the baseline. I carried forward the standardized dataset without feature drops.

Phase 4 — Outlier Detection

I applied column-wise Z-score outlier detection, removing rows from training data only where the absolute Z-score exceeded threshold T in at least m columns.

Attempt 1 — T=3, m=2: Removed 7 rows. $R^2 = 0.1516$. Marginal improvement over baseline.

Attempt 2 — T=2.5, m=2: Built on cleaned dataset from Attempt 1. Removed 116 additional rows. $R^2 = 0.1513$. Too aggressive — removing too many rows hurt performance.

Attempt 3 — T=3, m=3: Built on cleaned dataset from Attempt 1. Removed 0 rows. $R^2 = 0.1516$. No further outliers found with this stricter condition.

Results Summary

Stage	R^2 Score
Baseline	0.1515
Transform 1 (Min-Max)	0.1515
Transform 2 (Standardization)	0.1515
Transform 3 (Decimal Scaling)	0.1515
FS Attempt 1 (drop min_temp, max_temp, sunshine)	0.1480
FS Attempt 2 (drop min_temp, max_temp)	0.1489
FS Attempt 3 (drop snow_depth)	0.1515
Outlier Attempt 1 (T=3, m=2)	0.1516 *BEST*
Outlier Attempt 2 (T=2.5, m=2)	0.1513
Outlier Attempt 3 (T=3, m=3)	0.1516

Final $R^2 = 0.1516$ — achieved using Outlier Detection Attempt 1 (T=3, m=2), removing 7 outlier rows from the standardized training data.